

Framing Members and Openings

Naming the pieces in the order they stack, and the two questions about what happens when you cut a hole in a wall.

Sill	The 2x4 or 2x6 framing member laid horizontally on the concrete foundation and bolted down. It is the first piece of wood in the building.
Sole plate	The bottom horizontal member of a stud wall, sitting on the subfloor or the slab. Do not confuse it with the sill, which sits directly on the foundation.
Header	What is used to help support the wall and its load when an opening is made in a load-bearing wood framed wall. The load that used to run down through the missing studs is carried across the opening instead.
Rough opening	The measurement between the inside faces of the trimmers for a door. It is the framed hole, before the door and its frame go in.

YOU HAVE MISSED THIS

The 2x4 or 2x6 framing member placed horizontally on the concrete foundation and bolted down is the: **Sill.**

YOU HAVE MISSED THIS

When an opening is made in a load-bearing wood framed wall, a _____ is used to help support the wall and its load. **Header.**

YOU HAVE MISSED THIS

The measurement between the inside faces of the trimmers for a door is called: **The rough opening.**

The numbers here

Attic access rough framed opening — not less than

22 x 30 in

YOU HAVE MISSED THIS

The rough framed opening for attic access shall not be less than: **22 x 30 inches.**

Which way up a joist goes

YOU HAVE MISSED THIS

Floor joists should be installed:

Crown up.

A piece of lumber is never perfectly straight — it has a slight bow, and the high edge of that bow is the crown. Install it crown up and the load flattens it toward straight. Install it crown down and the load deepens the sag it already has.

Truss uplift

YOU HAVE MISSED THIS 2 TIMES

Truss uplift is typically caused by:

Differential humidity conditions in the attic.

The top chords of a truss sit up in the cold, dry attic air; the bottom chord is buried in insulation and stays warmer and damper. The two dry out at different rates, the bottom chord arches upward, and the ceiling lifts away from the interior partition walls in winter and settles back in summer. It is a moisture effect, not a structural failure — which is why the answer is humidity and not load, settlement or a defect.

Say it back

Cover the page. Answer each one out loud.

1. Sill against sole plate — which sits on what?
2. What goes in when you open a load-bearing wall?
3. What is a rough opening measured between?
4. Minimum attic access rough opening?
5. Which way up does a floor joist go, and what happens if it is reversed?
6. What causes truss uplift, and why does it come and go with the season?